

ML-2024 Assignment-2 (Naive Bayes)

Title: Comparison of Custom Naive Bayes and Sklearn Naive Bayes Models for Binary Classification

Group Members:

- Satish Jhanwer (G24AIT009@iitj.ac.in)
- Aditya Sharma (G24AIT011@iitj.ac.in)
- Jyothsna Sravanthi (G24AIT015@iitj.ac.in)

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Problem Statement:

We are tasked with implementing a binary classification module using the Ionosphere dataset. The goal is to classify the data points as either **good** or **bad**, based on radar signals, and compare two implementations: one from scratch and the other using Sklearn.

Dataset Overview

The **Ion Binary Classification Dataset** from the UCI Machine Learning Repository was used for this project. This dataset contains features derived from radar signals that classify whether a signal is **good** (Class 1) or **bad** (Class 0).

Both models were trained on **80% of the data** and tested on the remaining **20%** to evaluate their performance.

Naive Bayes from Scratch

We implemented a Gaussian Naive Bayes classifier from scratch using the following steps:

- **Calculate Mean and Variance:** For each class (good/bad), calculate the mean and variance for each feature.
- **Probability Calculation:** Compute the class-conditional probabilities using **Gaussian Probability Density**.
- **Posterior Calculation:** Use Bayes Theorem to compute posterior probabilities for each class and make a prediction based on the highest probability.
- **Evaluation:** Measure accuracy, precision, recall, and F1 score.

Naive Bayes Using Sklearn

We also used the GaussianNB classifier from the Sklearn library for comparison:

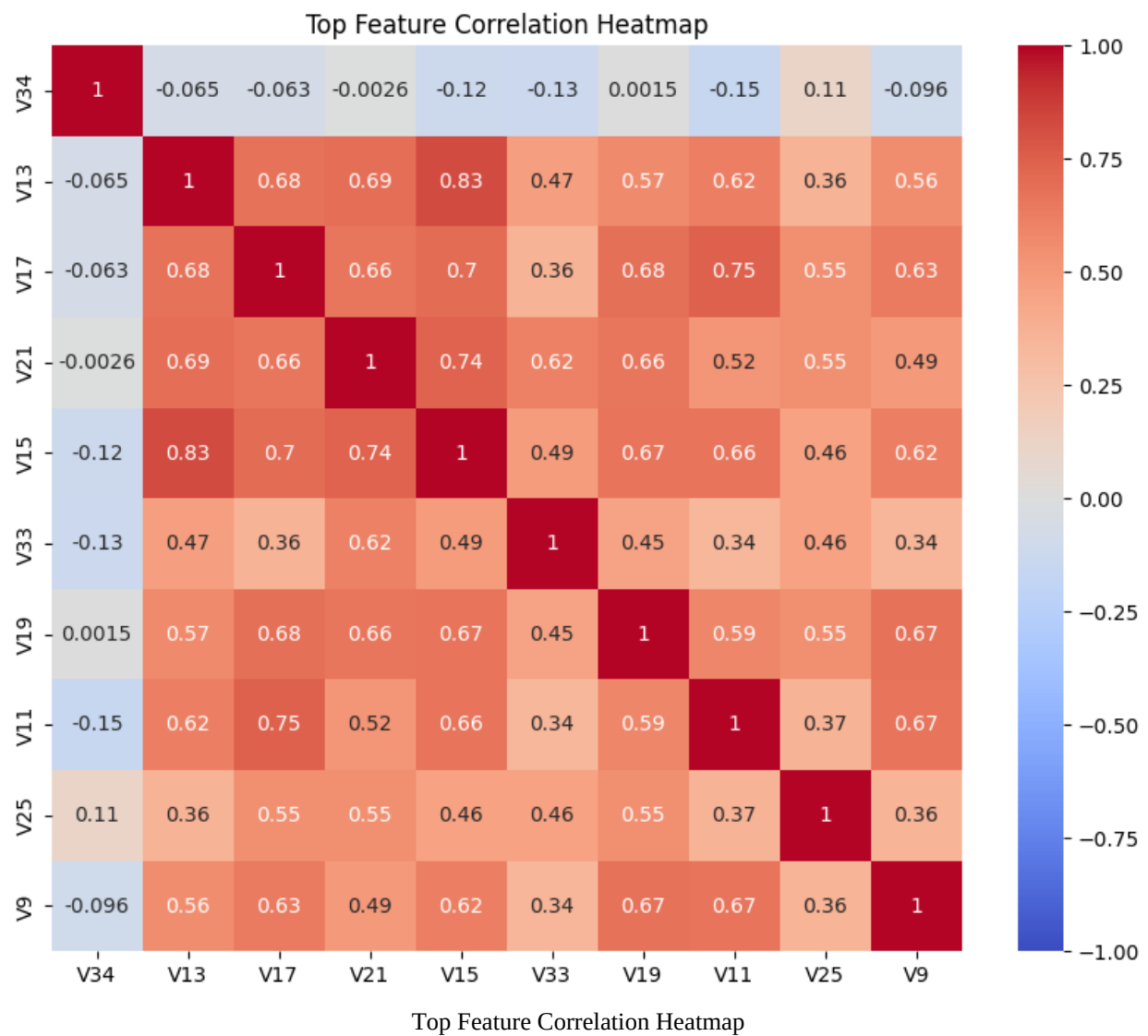
- **Model Training:** Trained the model using the training data split.
- **Evaluation:** Predicted on test data and calculated performance metrics.

Evaluation Metrics

The following metrics were used for performance evaluation:

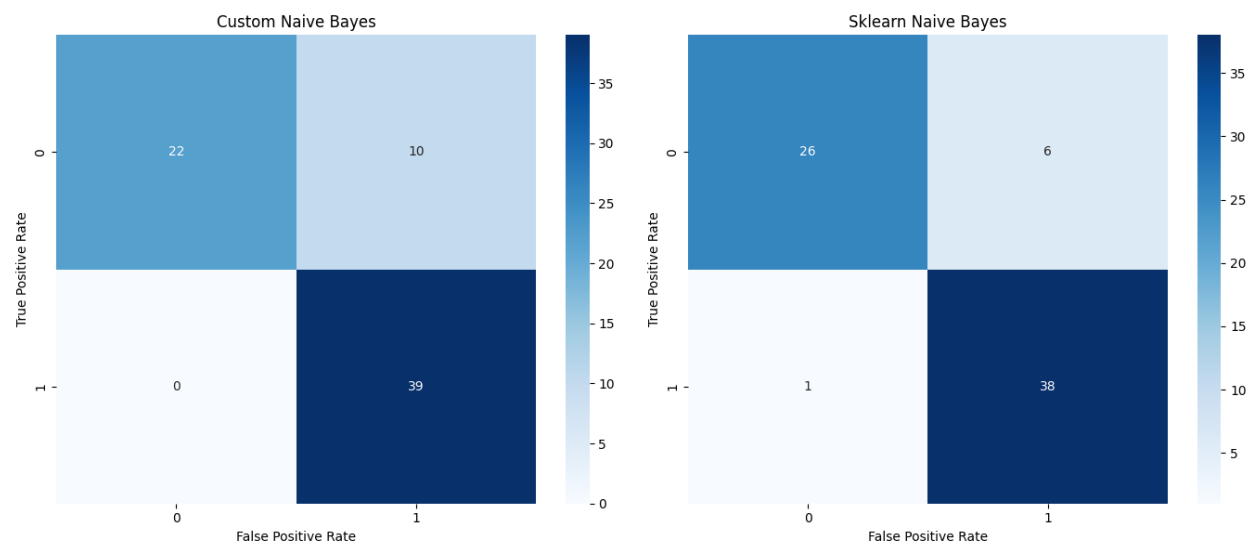
- **Accuracy:** The percentage of correctly classified instances.
- **Precision:** The percentage of correct positive predictions.
- **Recall:** The percentage of actual positives correctly classified.
- **F1 Score:** The harmonic mean of precision and recall.
- **Confusion Matrix:** A matrix to visualize misclassifications.

Feature Correlation Heatmap



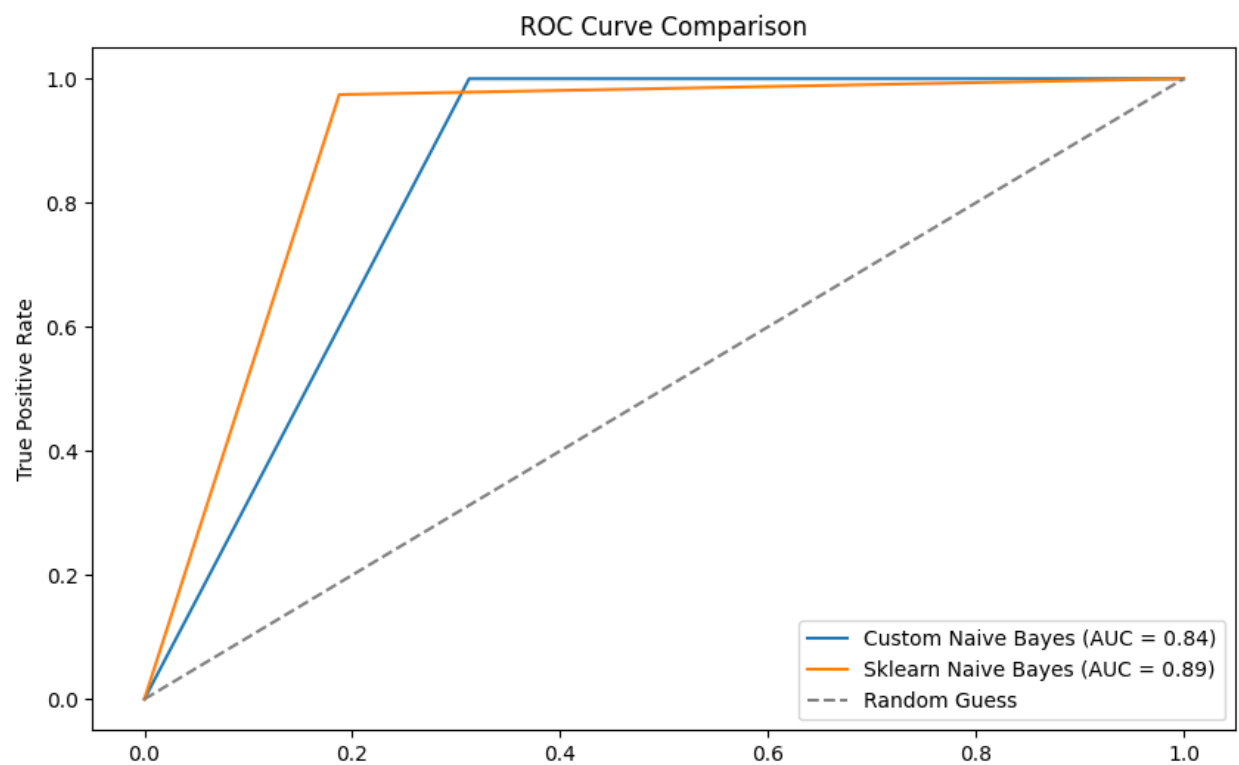
Results

Confusion Matrix Comparison



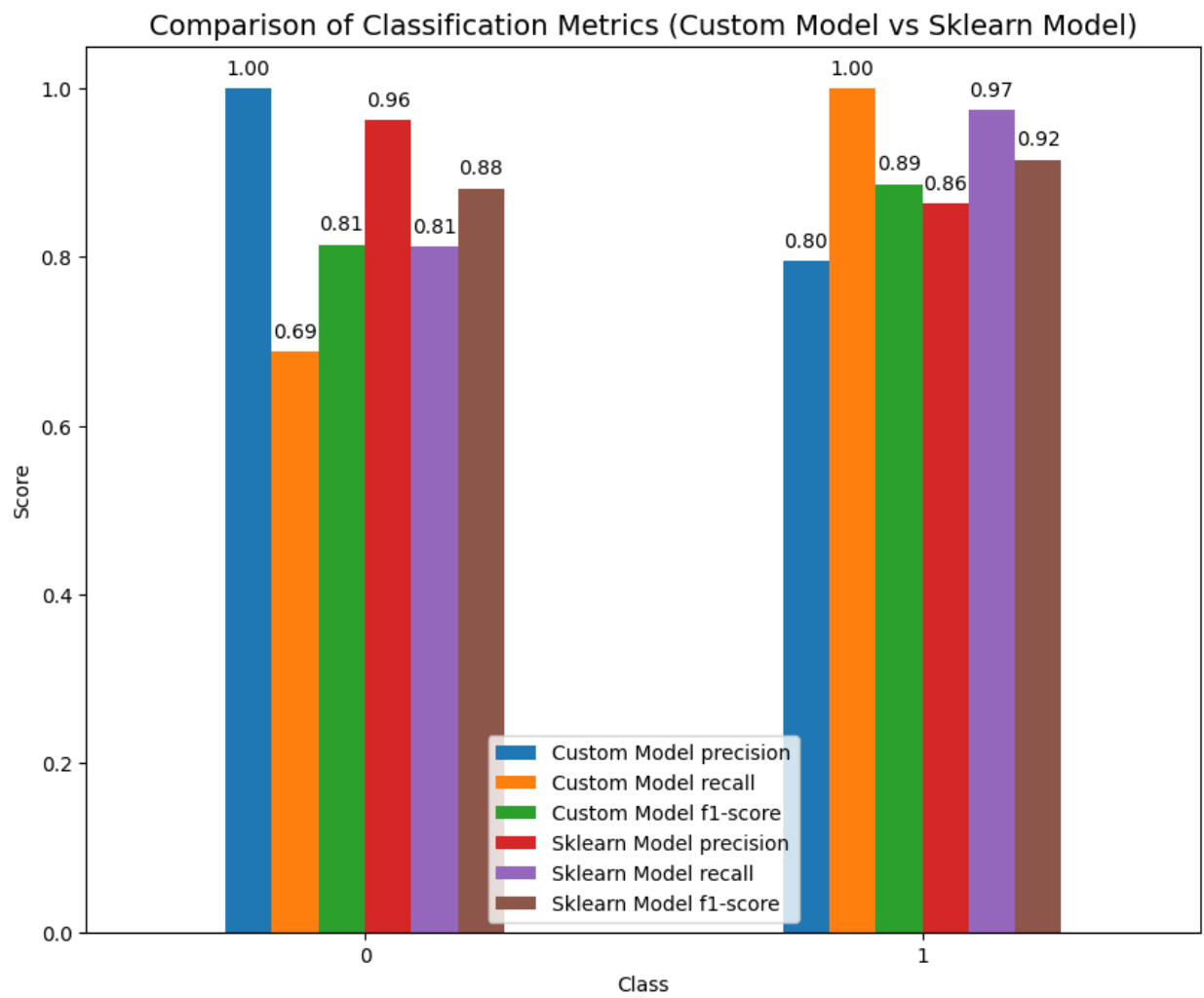
Confusion Matrix Comparison

ROC Curve Comparison



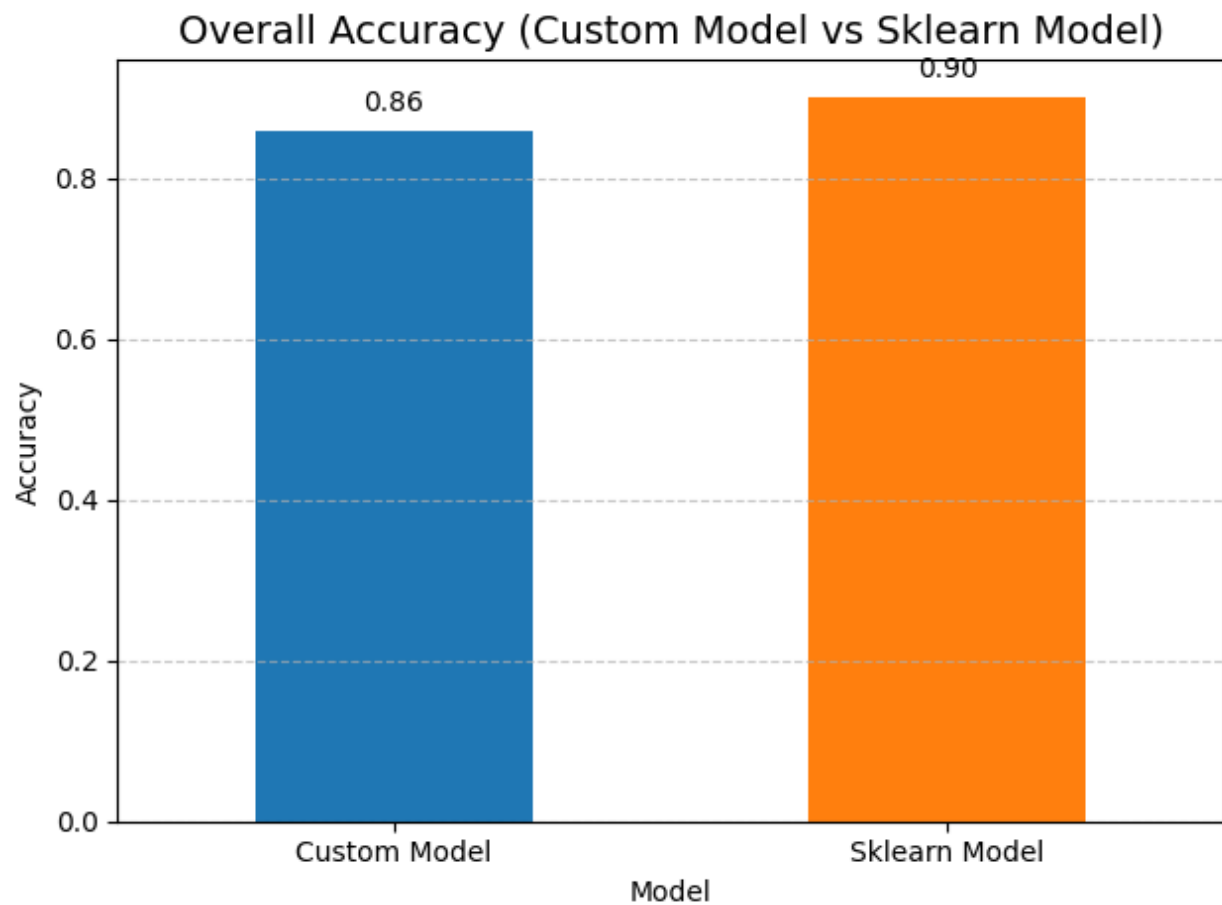
ROC Curve Comparison

Overall Result



Classification Comparison Report

Accuracy & Time Comparison



Accuracy Comparison

- Time for Naive Bayes from Scratch: 0.052051 seconds
- Time for Sklearn Naive Bayes: 0.006583 seconds

Conclusion

The Sklearn implementation is faster and more accurate, benefiting from optimized underlying libraries. However, our custom implementation gives insight into how Naive Bayes works at the foundational level.